

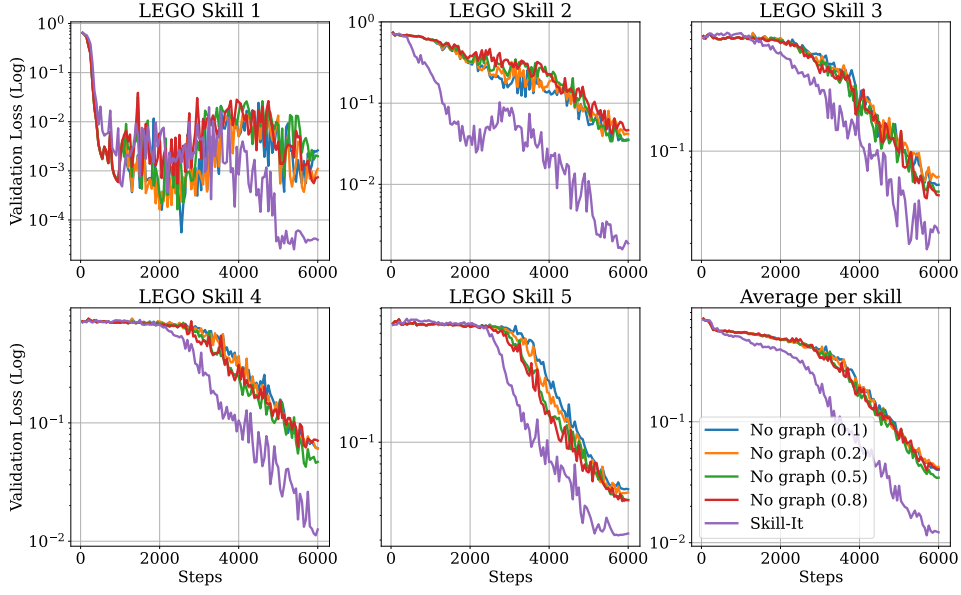


Figure 26: Comparison of SKILL-IT versus using the identity adjacency matrix (no skills graph) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the LEGO continual pre-training experiment. The latter does not capture the relationship between skills, and we find that SKILL-IT attains lower loss on all skills.

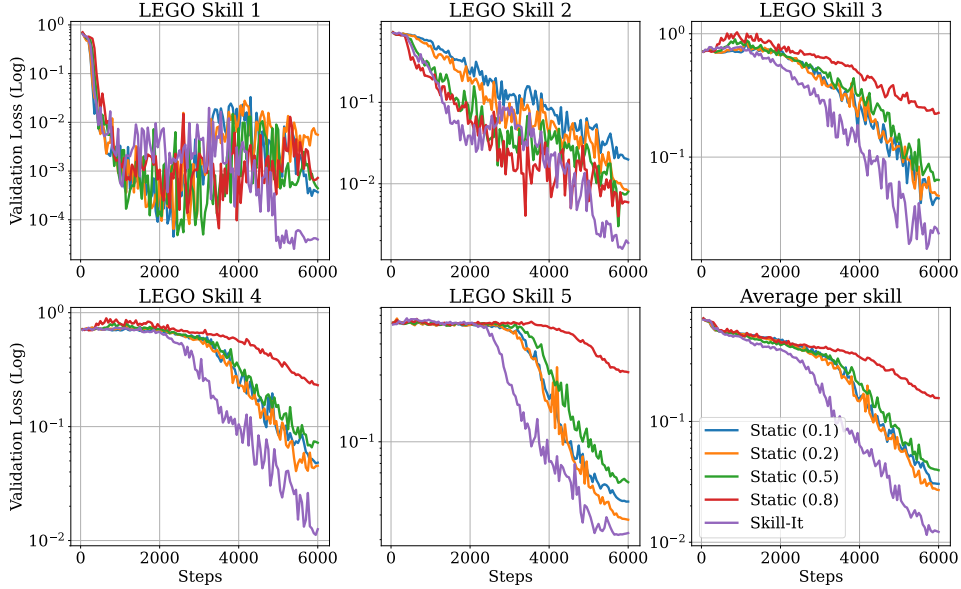


Figure 27: Comparison of SKILL-IT versus using static data selection ($T = 1$) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the LEGO continual pre-training experiment. While SKILL-IT eventually allocates more weights to skills 3, 4, 5, which have higher loss, the static approach is not able to do this. We find that SKILL-IT attains lower loss on all skills.

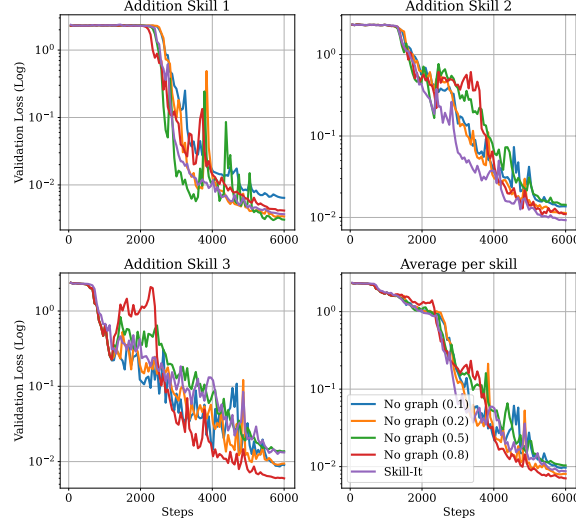


Figure 28: Comparison of SKILL-IT versus using the identity adjacency matrix (no skills graph) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the Addition continual pre-training experiment. The latter does not capture the relationship between skills, and we find that SKILL-IT attains lower loss on skill 2, but attains similar performance to methods that do not use the skills graph.

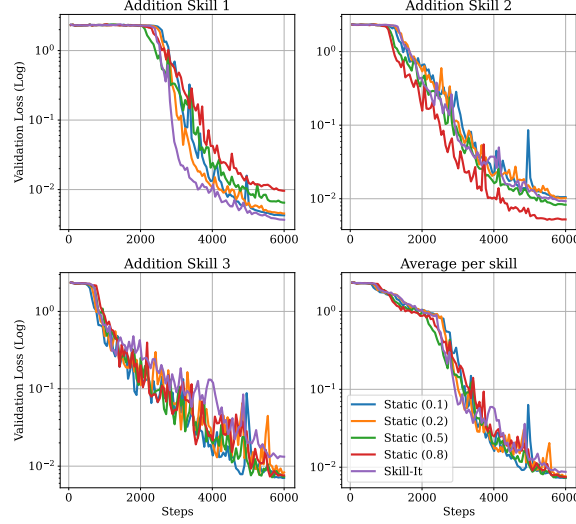


Figure 29: Comparison of SKILL-IT versus using static data selection ($T = 1$) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the Addition continual pre-training experiment. We find that SKILL-IT attains lower loss on skill 1, but attains similar performance to the static methods.

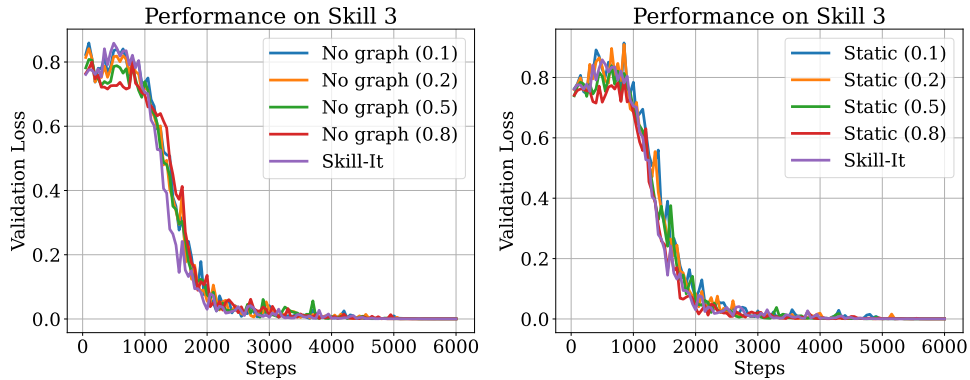


Figure 30: Comparison of SKILL-IT versus using no graph (left) and static data selection (right) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the LEGO fine-tuning experiment. All approaches have roughly the same loss trajectories, but SKILL-IT is slightly lower than using no graph.

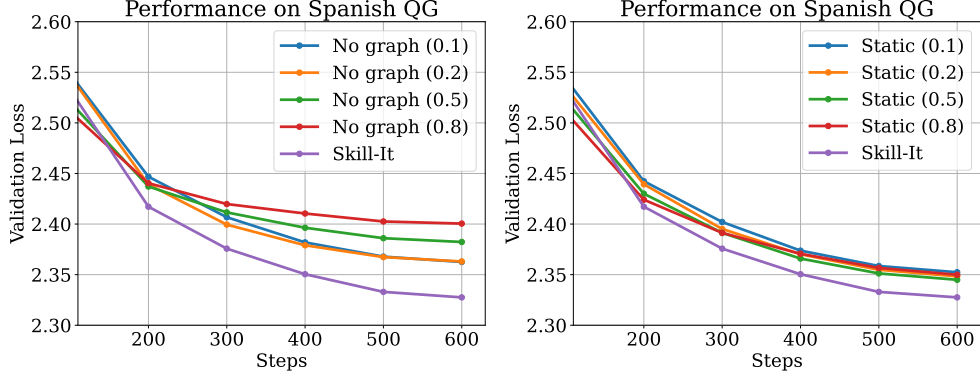


Figure 31: Comparison of SKILL-IT versus using no graph (left) and static data selection (right) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the Natural Instructions Spanish QG fine-tuning experiment. SKILL-IT attains lower validation loss than both no graph and static data selection.

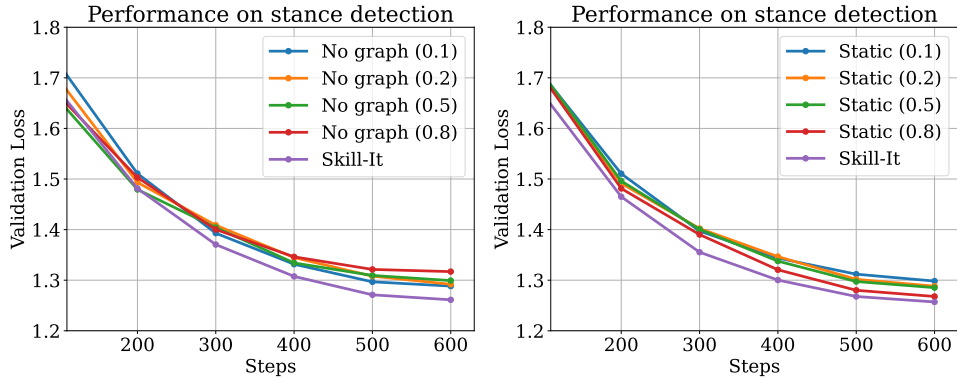


Figure 32: Comparison of SKILL-IT versus using no graph (left) and static data selection (right) with $\eta = 0.1, 0.2, 0.5, 0.8$ on the Natural Instructions stance detection fine-tuning experiment. SKILL-IT attains lower validation loss than both no graph and static data selection.